CYART

inquiry@cyart.io

www.cyart.io

Mobile Application Security Testing Report

Introduction

Mobile applications often handle sensitive user information such as credentials, location data, and personal messages. Because of this, it is important to analyze applications for potential vulnerabilities before installing or using them. This report describes the static security analysis of an Android application using an online malware analysis tool.

Objective

The objective of this task is to perform static security analysis on an Android application and identify potential security risks.

Tools Used

- Kali Linux
- VirusTotal

Target Application

Application Name: InsecureBankv2

File Type: Android APK

Purpose: Security testing application

The application InsecureBankv2 is intentionally designed to contain security vulnerabilities so that students and security researchers can practice mobile application testing.

Methodology

The following steps were performed during the analysis:

1. Downloaded the vulnerable Android APK file.
2. Uploaded the APK file to the VirusTotal platform.
3. VirusTotal scanned the application using multiple security engines.
4. The results were analyzed to identify suspicious behaviors and vulnerabilities.

Analysis Results

The VirusTotal scan revealed that several antivirus engines flagged the application as malicious.

Key findings:

- 17 out of 67 security vendors detected the file as malicious.
- The application includes suspicious behaviors such as sending SMS messages and accessing telephony services.
- The application also shows signs of code obfuscation and reflection.

Detected Permissions and Behaviors

The analysis revealed several potentially dangerous capabilities:

- Access to telephony services
- Ability to send SMS messages
- Clipboard access
- GPS location checks
- Use of reflection in code

Such permissions could be abused in malicious applications to perform unauthorized actions on a user's device.

Conclusion

The static analysis demonstrates that the InsecureBankv2 application contains several security issues and potentially dangerous behaviors. However, this is expected because the application is intentionally vulnerable for security training purposes.

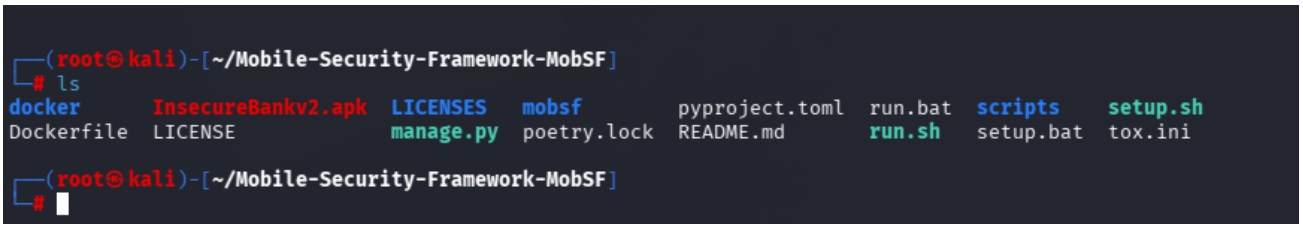
Using tools like VirusTotal helps security analysts quickly identify suspicious applications and analyze potential threats before deployment or installation.

Screenshots

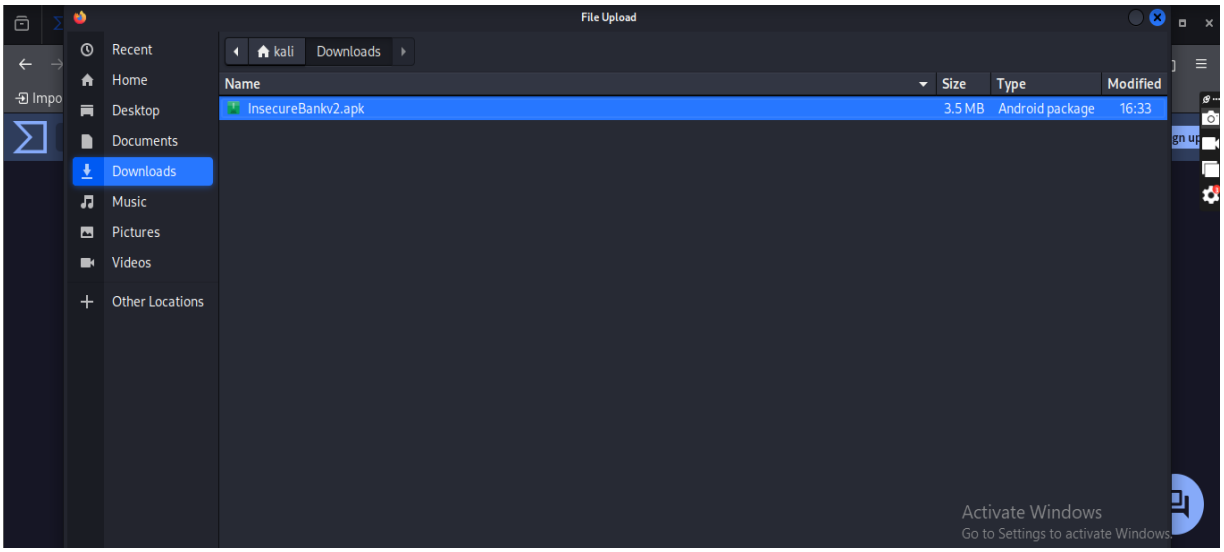
The following screenshots were captured during the analysis process:

1. APK download in Kali Linux

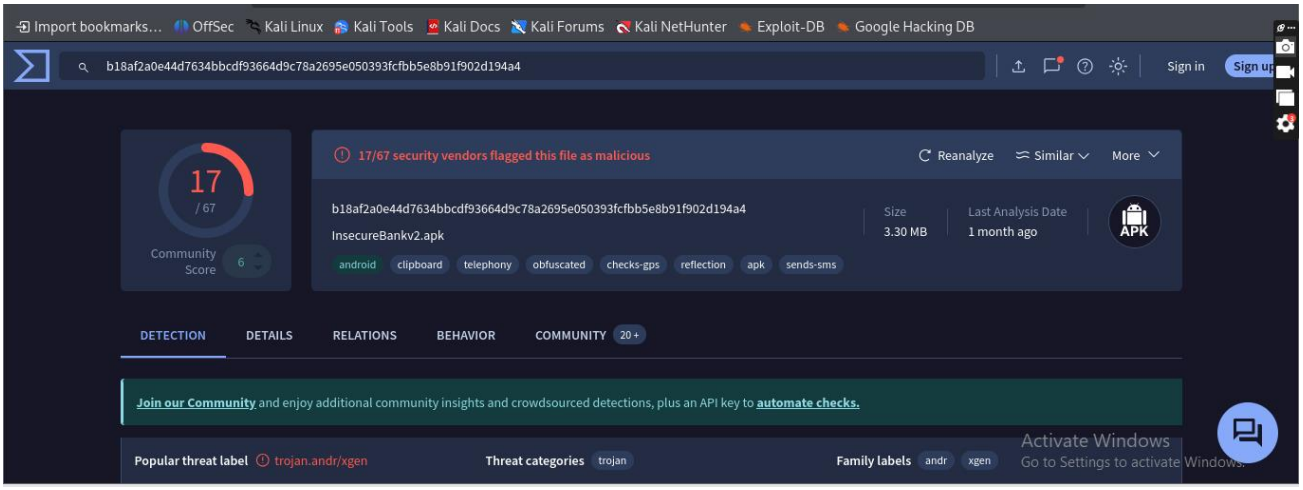
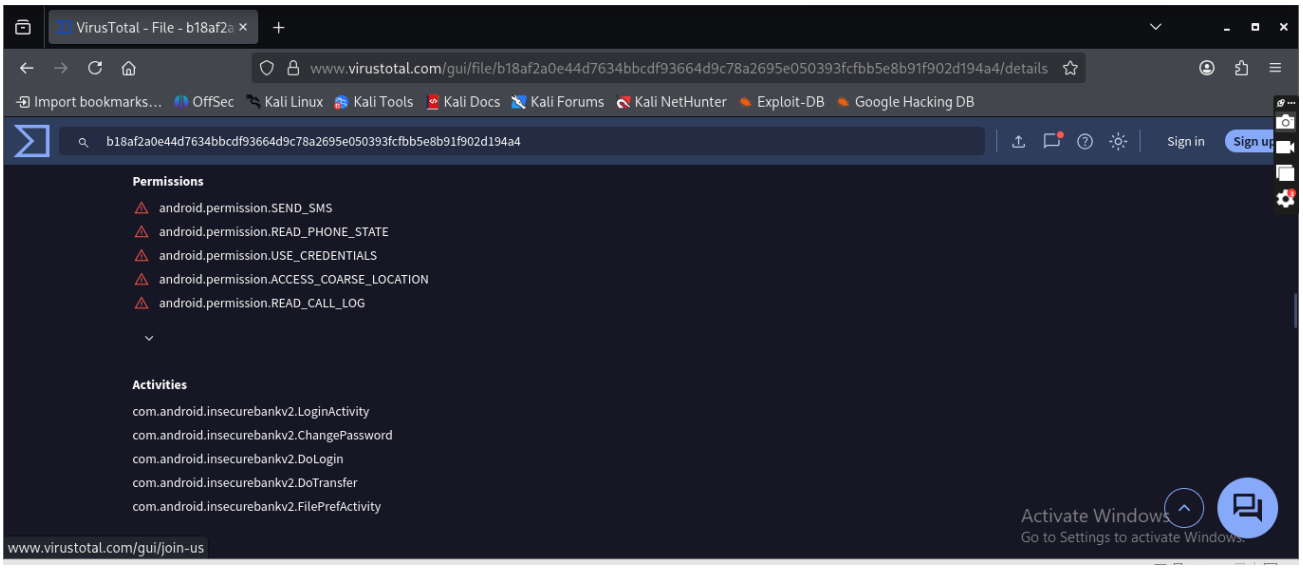
2. APK file verification



3. APK upload to VirusTotal



4. VirusTotal detection results



5. Permissions and details page